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| **MATEMATIKA II** **2.EBALUAKETA – PARTZIALA** **Data:2025-12-18**<br>**IZEN-ABIZENAK:____________________________ TALDEA: _____**<br><br>**Planteamenduak egokiak izan behar dira, kontzeptuak, hizkuntza eta notazio zientifikoa zuzenak izan behar dira eta egindako urratsen azalpen garbia eta aurkezpena txukuna izan beharko dira.** | |

**1. Kalkulatu hurrengo integral mugagabeak, berehalako integralak bezala edo ordezkapen metodoa erabiliz:(2p)**

a) $\int \frac{\sqrt{3}\ dx}{3+9x^2} = \sqrt{3}\int \frac{dx}{3(1+3x^2)} = \frac{\sqrt{3}}{3\sqrt{3}}\int \frac{\sqrt{3}\,dx}{1+(\sqrt{3}x)^2} = \boxed{\frac{1}{3}\ arctg(\sqrt{3}x)+k}$

$\boxed{\int f^n \cdot f' dx = \frac{f^{n+1}}{n+1}}$ 1.MODUA $I = \int (2x+1)^{-1/3} dx = \frac{1}{2}\int 2(2x+1)^{-1/3} dx =$

b) $\int \frac{dx}{\sqrt[3]{2x+1}} = \quad = \frac{1}{2}\cdot\frac{(2x+1)^{-1/3+1}}{-1/3+1} = \frac{\sqrt[3]{(2x+1)^2}}{2\cdot 2/3} = \boxed{\frac{3}{4}\sqrt[3]{(2x+1)^2}+k}$

2.MODUA (ordezkapen metodoa)

$t = \sqrt[3]{2x+1}$
$t^3 = 2x+1$
$3t^2dt = 2dx$
$dx = \frac{3t^2}{2}dt$

$I = \int \frac{3t^2dt}{2t} = \int \frac{3}{2}t\,dt = \frac{3}{2}\frac{t^2}{2} =$

$= \boxed{\frac{3}{4}\sqrt[3]{(2x+1)^2}+k}$

d) $\int \frac{\sin x}{1+\cos x}dx = \boxed{-Ln\,|1+\cos x|+k}$

$t = 1+\cos x$
$dt = -\sin x\,dx$

2.MODUA (ORDEZKAPENA)

$I = \int \frac{-dt}{t} = -Ln|t| = -Ln|1+\cos x|+k$

**2. Kalkulatu ondorengo integral mugagabea <u>ordezkapen-metodoa</u> erabiliz: (1p)**

$\int \frac{\sqrt{x}}{1+x}\cdot dx = \int \frac{t\cdot 2t\,dt}{1+t^2} = \int \frac{2t^2}{1+t^2}dt = 2\int \frac{t^2}{1+t^2}dt$

$t = \sqrt{x}$
$t^2 = x$
$2t\,dt = dx$

$= 2\int \left(1+\frac{-1}{1+t^2}\right)dx =$

$= 2\cdot(t - arctg\,t) =$

$t^2 \div (1+t^2) = 1$
$-t^2-1$
$\overline{\quad -1 \quad}$

$= \boxed{2\cdot(\sqrt{x} - arctg\sqrt{x})+k.}$

**3.** Kalkulatu ondorengo integral mugagabeak: **2 + 1,5 =3,5**

$$\int \frac{2x^2+1}{x^3+2x^2+x}dx$$

$x^3+2x^2+x = x(x^2+2x+1) = x(x+1)^2$

Erroak $\begin{cases} x=0 \\ x=-1 \text{ bikoitza} \end{cases}$

$$\frac{2x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

$$2x^2+1 = A(x+1)^2 + Bx(x+1) + Cx.$$

$x=0 \longrightarrow 1 = A + \cancel{B\cdot 0} + \cancel{C0} \longrightarrow \boxed{A=1}$

$x=-1 \longrightarrow 3 = \cancel{A0} + \cancel{B\cdot 0} - C \longrightarrow \boxed{C=-3}$.

$x=1 \longrightarrow 3 = 4A + 2B + C$

$3 = 4 + 2B - 3$

$2 = 2B \longrightarrow \boxed{B=1}$

$$I = \int \frac{2x^2+1}{x(x+1)^2}dx = \int \left( \frac{1}{x} + \frac{1}{x+1} + \underbrace{\frac{-3}{(x+1)^2}}_{*} \right) dx$$

$$= \ln|x| + \ln|x+1| + 3\frac{1}{x+1} + k$$

$$= \boxed{\ln|x(x+1)| + \frac{3}{x+1} + k}$$

$*$ $\int \frac{1}{(x+1)^2}dx = \int (x+1)^{-2}dx = \frac{(x+1)^{-1}}{-1} = \frac{-1}{x+1}$

$I = \int \frac{2x^2 - 9x - 5}{x^2 - 3x} dx$ Deg $P(x) \geq$ Deg $Q(x) \rightarrow$ zatiketa.

$$\begin{array}{l|l} 2x^2 - 9x - 5 & x^2 - 3x \\ -2x^2 + 6x & 2 \\ \hline / \; -3x - 5 & \end{array}$$

$I = \int 2 + \frac{-3x-5}{x^2-3x} dx =$

$= \int \left(2 - \frac{3x+5}{x^2-3x}\right) dx$

Erro simpleak $x=0$, $x=3$

$x^2 - 3x = x(x-3)$

$\frac{3x+5}{x(x-3)} = \frac{A}{x} + \frac{B}{x-3}$

$3x+5 = A(x-3) + Bx$

$x = 0 \rightarrow 5 = -3A$

$x = 3 \rightarrow 14 = 3B.$

$\boxed{A = -5/3 \quad B = 14/3}$

$I = \int \left(2 - \left(\frac{-5/3}{x} + \frac{14/3}{x-3}\right)\right) dx$

$\boxed{I = 2x + \frac{5}{3} Ln|x| - \frac{14}{3} Ln|x-3| + K}$

4. **Kalkulatu integral mugagabe hauek zatikako metodoa erabiliz** 1,5+2 =3,5 p

$\int \frac{x}{e^{4x}} dx = \int \underset{P}{x} \cdot \underset{E}{e^{-4x}} dx.$

$\left\{ \begin{array}{l} u = x \rightarrow du = dx \\ dv = e^{-4x} dx \rightarrow v = \frac{1}{-4} \int -4 e^{-4x} dx = -\frac{1}{4} e^{-4x} \end{array} \right.$

$\int u \, dv = u \cdot v - \int v \cdot du$

$\int x \cdot e^{-4x} dx = x \cdot \left(-\frac{1}{4} e^{-4x}\right) - \int -\frac{1}{4} e^{-4x} dx =$

$= -\frac{1}{4} x e^{-4x} + \frac{1}{4} \frac{1}{-4} \int -4 e^{-4x} dx =$

$= -\frac{1}{4} x e^{-4x} - \frac{1}{16} e^{-4x} + K$

$\boxed{I = \frac{e^{-4x}}{4}\left(-x - \frac{1}{4}\right) + K.}$

$I = \int (x^2 - 2x) \cdot \sin(2x)\, dx$ (P) (S)

$I = \begin{cases} u = x^2 - 2x \longrightarrow du = (2x-2)\,dx \\ dv = \sin(2x)\,dx \longrightarrow \end{cases}$

$v = \frac{1}{-2}\int -2\sin(2x)\,dx = -\frac{1}{2}\cos(2x)$

$\int u\,dv = uv - \int v\,du$

$I = (x^2-2x)\cdot\left(-\frac{1}{2}\right)\cdot\cos(2x) - \int\left(-\frac{1}{2}\right)\cdot\cos(2x)\cdot(2x-2)\cdot dx$

$= -\frac{1}{2}\cdot(x^2-2x)\cdot\cos(2x) + \int \frac{1}{\cancel{2}}\cdot\cancel{2}\cdot(x-1)\cdot\cos(2x)\cdot dx$

$= -\frac{1}{2}\cdot(x^2-2x)\cos(2x) + \underbrace{\int (x-1)\cdot\cos(2x)\,dx}_{I_1}$ (P) (S)

$I_1 = \begin{cases} u = x-1 \longrightarrow du = dx \\ dv = \cos 2x\,dx \longrightarrow v = \frac{1}{2}\int 2\cos(2x)\,dx \end{cases}$

$= 1/2 \cdot \sin(2x)$

$I_1 = (x-1)\cdot\frac{1}{2}\cdot\sin(2x) - \int\frac{1}{2}\cdot\sin(2x)\cdot dx$

$= (x-1)\,\frac{1}{2}\cdot\sin(2x) - \frac{1}{2}\,\frac{1}{-2}\int -2\sin(2x)\,dx =$

$= \frac{1}{2}\cdot(x-1)\sin(2x) + \frac{1}{4}\cdot\cos(2x)$

$I = -\frac{1}{2}\cdot(x^2-2x)\cdot\cos(2x) + I_1$

$I = -\frac{1}{2}(x^2-2x)\cdot\cos(2x) + \frac{1}{2}\cdot(x-1)\cdot\sin(2x) + \frac{1}{4}\cdot\cos(2x) + K$

$$\boxed{I = \frac{1}{2}\left[\cos(2x)\left(-x^2+2x+\frac{1}{2}\right) + \sin(2x)\cdot(x-1)\right] + K}$$